

Product Requirements Document (PRD)

Inventory Barcode Generator with Asset Tracking — Working Prototype

Version: 1.0 (Prototype MVP)

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Status: Draft for Review

Prepared for: Product & Engineering Team

1. Document Purpose

This PRD defines the requirements for a **working prototype** (not the full product) of the Inventory Barcode Generator with Asset Tracking system. The prototype is designed to validate core assumptions with four real user personas identified through LinkedIn profile analysis and market research.

Prototype Scope: 6-week build, 3-week pilot with 2-3 beta users per persona segment

Success Criteria: 80% task completion rate, <5 minute setup time, NPS 40

2. Executive Summary

2.1 Product Vision

Enable small businesses (100-5,000 SKUs) transitioning from manual/spreadsheet inventory tracking to their first barcode system — without expensive software, dedicated hardware, or data migration.

2.2 Prototype Hypothesis

If we provide a three-step setup (upload → generate → print) with Google Sheets as the persistent data layer and smartphone-based scanning, then small business owners will adopt barcode inventory management within 30 minutes without IT support.

2.3 Target Segments (from Persona Analysis)

Segment	Persona	Primary Pain	Decision Speed	Price Sensitivity
Small Retail	Koteswara Rao (KD)	Multi-category chaos	Fast	Very High
Equipment Rental	Indrajit Paul (IP)	Asset visibility across sites	Slow	Low (enterprise)
Startup Operations	Samrudhi Sudhakar (SS)	Rapid scaling breaking systems	Medium	Low
Medical Administration	Andrea Bissonnette (AB)	Compliance + audit readiness	Very Slow	Medium

2.4 Prototype vs. Full Product Scope

Feature Area	Prototype (MVP)	Full Product (Post-Validation)
Barcode Formats	Code 128 + QR Code	Code 128, QR, Data Matrix, UPC/EAN
Label Formats	PDF (A4/Letter) + PNG (thermal)	PDF, PNG, ZPL, DYMO, Brother
Sync Backend	Direct Google Sheets API	Hybrid backend + Redis queue
User Roles	Single user + read-only sharing	3-tier RBAC (Admin/Manager/Operator)
Offline Mode	Basic queued sync	Full offline with conflict resolution
Integrations	Google Sheets only	Airtable, Firebase, Slack, QuickBooks
Analytics	Basic turnover report	Full dashboard with predictive alerts

3. User Personas & Prototype Scenarios

3.1 Persona 1: Koteswara Rao Devarasetty (KD)

Segment: Small Retail Owner
Company: BSKB Communications, Guntur, India
Scenario: Track 500+ SKUs across mobiles, TVs, laptops, ACs, accessories, and repair parts

Prototype Success Criteria for KD:

- Upload existing Google Sheet with 500+ rows in <2 minutes
- Generate barcodes for all items in <10 minutes
- First inventory count completed in <1 hour (vs. 4 hours manual)
- Discover at least 5 discrepancies on first scan (validates value)

Critical Path:

Google Workspace Marketplace discovery → SSO signup → Sheet upload → Barcode generation (batch) → PDF label print → Mobile PWA scan → Discrepancy discovery → Google Sheets sync verification

3.2 Persona 2: Indrajit Paul (IP)

Segment: Equipment Rental / Enterprise Asset Management
Company: Larsen & Toubro, Chennai, India
Scenario: Pilot digital asset tracking for 50 rental equipment units across 3 project sites

Prototype Success Criteria for IP:

- Upload CSV extract from SAP/ERP in <5 minutes
- Configure 4 custom fields (Asset ID, Site, Rental Status, Maintenance Due)
- Deploy to 3 sites with 90%+ daily scan compliance within 1 week
- Generate first utilization report showing >70% actual vs. estimated

Critical Path:

Industry conference demo → Pilot approval → SAP CSV upload →
Custom field mapping → ZPL/thermal label generation → Multi-site deployment →
Daily scan dashboard → Weekly utilization report

3.3 Persona 3: Samrudhi Sudhakar (SS)

Segment: Startup Operations / Demo Unit Tracking

Company: upliance.ai, Bengaluru, India

Scenario: Track 45 demo units across review, event, internal, and repair categories

Prototype Success Criteria for SS:

- Create 5 custom categories with auto due-dates in <10 minutes
- QR Code generation (brand-aligned, modern aesthetic)
- 100% team adoption (5+ members scanning within 48 hours)
- Zero lost demo units during 4-week pilot

Critical Path:

Product Hunt discovery → Slack team vote → Google SSO →
Demo unit spreadsheet upload → Category/due-date configuration →
QR label generation (Brother QL-800) → Team labeling party →
Check-out/check-in workflow → Weekly founder report

3.4 Persona 4: Andrea Bissonnette (AB)

Segment: Medical Office / Vaccine Inventory

Company: Carleton University, Ottawa, Canada

Scenario: Track 23 vaccine types, 150+ lots, with expiration alerts and audit compliance

Prototype Success Criteria for AB:

- Upload vaccine inventory with lot numbers and expiry dates in <5 minutes
- Configure 3-tier alert system (30/14/0 days) with color coding
- Pass mock audit with 100% lot traceability and timestamp verification
- Train 8 nurses to scan-before-administer in <20 minutes

Critical Path:

Health conference session → IT security review (3 weeks) →
Vaccine inventory upload → Expiration alert configuration →
Role-based access (Admin/Manager/Operator) → Small label generation (1"×0.5") →
Clinical staff training → Flu clinic 312-scan event → Audit report generation

4. Functional Requirements

4.1 Core Feature: Three-Step Setup

FR-1.1: Spreadsheet Upload

ID	Requirement	Priority	Acceptance Criteria
FR-1.1.1	Support Google Sheets direct import via API	P0	OAuth2 connection, <30s for 1000 rows
FR-1.1.2	Support CSV/Excel file upload	P0	Drag-drop, auto-detect headers, preview mapping
FR-1.1.3	Auto-detect column types (name, SKU, qty, location, category)	P1	80% accuracy on standard headers
FR-1.1.4	Manual column mapping for non-standard headers	P1	Dropdown mapping interface, save template
FR-1.1.5	Validate data integrity (duplicate SKUs, negative quantities)	P1	Highlight errors, suggest fixes

FR-1.2: Barcode Generation

ID	Requirement	Priority	Acceptance Criteria
FR-1.2.1	Generate Code 128 barcodes with custom prefix	P0	Prefix validation, sequential numbering
FR-1.2.2	Generate QR Code as secondary format	P0	Toggle between formats, same data payload
FR-1.2.3	Batch generation for 500+ items in <10 minutes	P0	Progress indicator, background processing
FR-1.2.4	Support custom fields in barcode payload (up to 4 fields)	P1	Field selector, payload preview
FR-1.2.5	Generate barcode reference sheet (index page)	P2	Printable A4/Letter summary

FR-1.3: Label Printing

ID	Requirement	Priority	Acceptance Criteria
FR-1.3.1	PDF output for standard printers (A4/Letter)	P0	Avery 5160 template, customizable layout
FR-1.3.2	PNG output for thermal printers (300 DPI)	P1	2" ×1" , 4" ×6" presets, monochrome
FR-1.3.3	Label layout editor (product name, SKU, barcode, location)	P1	Drag-drop fields, font size control

Table 5 – continued

ID	Requirement	Priority	Acceptance Criteria
FR-1.3.4	Print calibration wizard for alignment	P2	Test page, margin adjustment, save profile

4.2 Core Feature: Mobile PWA Scanner

FR-2.1: Scanning Capability

ID	Requirement	Priority	Acceptance Criteria
FR-2.1.1	Scan Code 128 and QR Code using smartphone camera	P0	<2s scan time, 95% accuracy in good light
FR-2.1.2	Auto-focus and torch/flash control	P0	Toggle button, auto-enable in low light
FR-2.1.3	Manual barcode entry fallback with validation	P1	Prefix check, format validation, error feedback
FR-2.1.4	Batch scan mode (continuous scanning without confirmation)	P1	Toggle mode, audio confirmation, count display

FR-2.2: Inventory Operations

ID	Requirement	Priority	Acceptance Criteria
FR-2.2.1	Check-in/check-out with timestamp and user ID	P0	One-tap operation, auto-populate fields
FR-2.2.2	Location transfer with from/to logging	P0	Dropdown locations, scan confirmation
FR-2.2.3	Quantity adjustment (increment/decrement/set)	P0	Numeric input, reason code, audit trail
FR-2.2.4	View item details (name, current qty, location, history)	P1	Expandable card, last 5 transactions

FR-2.3: Offline & Sync

ID	Requirement	Priority	Acceptance Criteria
FR-2.3.1	Queue scans locally when offline (IndexedDB)	P0	Store 500+ scans, visual offline indicator
FR-2.3.2	Auto-sync when connection restored	P0	Background sync, conflict alert, retry logic
FR-2.3.3	Sync status dashboard (pending/complete/failed)	P1	Count badge, retry button, error details

4.3 Core Feature: Google Sheets Integration

FR-3.1: Real-Time Sync

ID	Requirement	Priority	Acceptance Criteria
FR-3.1.1	Two-way sync: app writes to Sheet, Sheet changes reflect in app	P0	<30s latency, conflict detection
FR-3.1.2	Add system columns (barcode, last_scan_date, current_location)	P0	Auto-append, preserve existing data
FR-3.1.3	Handle Google Sheets API rate limits (300 req/min)	P0	Batch writes (50 rows), exponential backoff
FR-3.1.4	Data validation and error handling	P1	Highlight sync failures, suggest fixes

FR-3.2: Sheet Management

ID	Requirement	Priority	Acceptance Criteria
FR-3.2.1	Create new inventory Sheet from template	P1	Pre-populated headers, example row
FR-3.2.2	Connect to existing Sheet (read-write)	P0	OAuth consent, scope verification
FR-3.2.3	Disconnect and data portability (export CSV)	P1	Full export, no data lock-in

4.4 Prototype-Specific Features (Persona-Driven)

FR-4.1: For KD (Small Retail) — Multi-Category Support

ID	Requirement	Priority	Acceptance Criteria
FR-4.1.1	Category-based barcode prefix auto-assignment	P1	Mobiles=MOB, TVs=TV, ACs=AC, etc.
FR-4.1.2	Category-specific reorder point alerts	P2	Default thresholds, customizable per category
FR-4.1.3	Service parts tracking (repair inventory)	P2	Separate workflow, consumption logging

FR-4.2: For IP (Enterprise Pilot) — Custom Fields & Reporting

ID	Requirement	Priority	Acceptance Criteria
FR-4.2.1	Up to 10 custom fields per item	P1	Text, number, date, dropdown types
FR-4.2.2	Custom field mapping from CSV upload	P1	Auto-detect + manual override
FR-4.2.3	Basic utilization report (scan frequency by asset)	P2	Weekly email, CSV export
FR-4.2.4	Maintenance due date tracking	P2	Date field, 7-day advance alert

FR-4.3: For SS (Startup) — Team & Event Workflows

ID	Requirement	Priority	Acceptance Criteria
FR-4.3.1	Multi-user scan with user attribution	P1	Google account login, scan log shows user
FR-4.3.2	Check-out with automatic due-date	P1	Configurable default (7 days review, 1 day event)
FR-4.3.3	Overdue alert (email + in-app)	P1	Daily digest, individual reminders
FR-4.3.4	Location transfer for events	P1	“Event” location type, auto-return date

FR-4.4: For AB (Medical) — Expiration & Compliance

ID	Requirement	Priority	Acceptance Criteria
FR-4.4.1	Expiration date field with 3-tier alerts	P1	30 days (green), 14 days (amber), 0 days (red)
FR-4.4.2	Lot number tracking and traceability	P1	Unique lot field, scan shows lot + expiry
FR-4.4.3	“Use first” tray flag for near-expiry items	P2	Auto-flag, visual indicator in app
FR-4.4.4	Audit trail report (user, timestamp, action, lot)	P2	Filterable by date range, CSV/PDF export
FR-4.4.5	Scan-before-administer confirmation	P2	Voice confirmation, wrong item warning

5. Non-Functional Requirements

5.1 Performance

ID	Requirement	Target	Measurement
NFR-1	Page load time (initial)	<3s	Lighthouse Performance Score 90

Table 15 – continued

ID	Requirement	Target	Measurement
NFR-2	Barcode generation (500 items)	<10 min	Server-side processing, progress API
NFR-3	Scan-to-sync latency (online)	<5s	End-to-end with Google Sheets API
NFR-4	Offline scan queue capacity	500 scans	IndexedDB storage limit
NFR-5	Concurrent users (prototype)	10 users	Load testing with Artillery/K6

5.2 Security & Compliance

ID	Requirement	Priority	Notes
NFR-6	Google OAuth 2.0 with minimal scopes	P0	<code>drive.file</code> only, no broad Drive access
NFR-7	Data encryption in transit (TLS 1.3)	P0	Certificate pinning for API
NFR-8	No persistent storage of user inventory data	P0	All data in user' s Google Drive
NFR-9	HIPAA/FIPPA compliance documentation	P2	For AB segment, pre-built compliance pack
NFR-10	Audit logging (admin actions)	P1	90-day retention, exportable

5.3 Usability

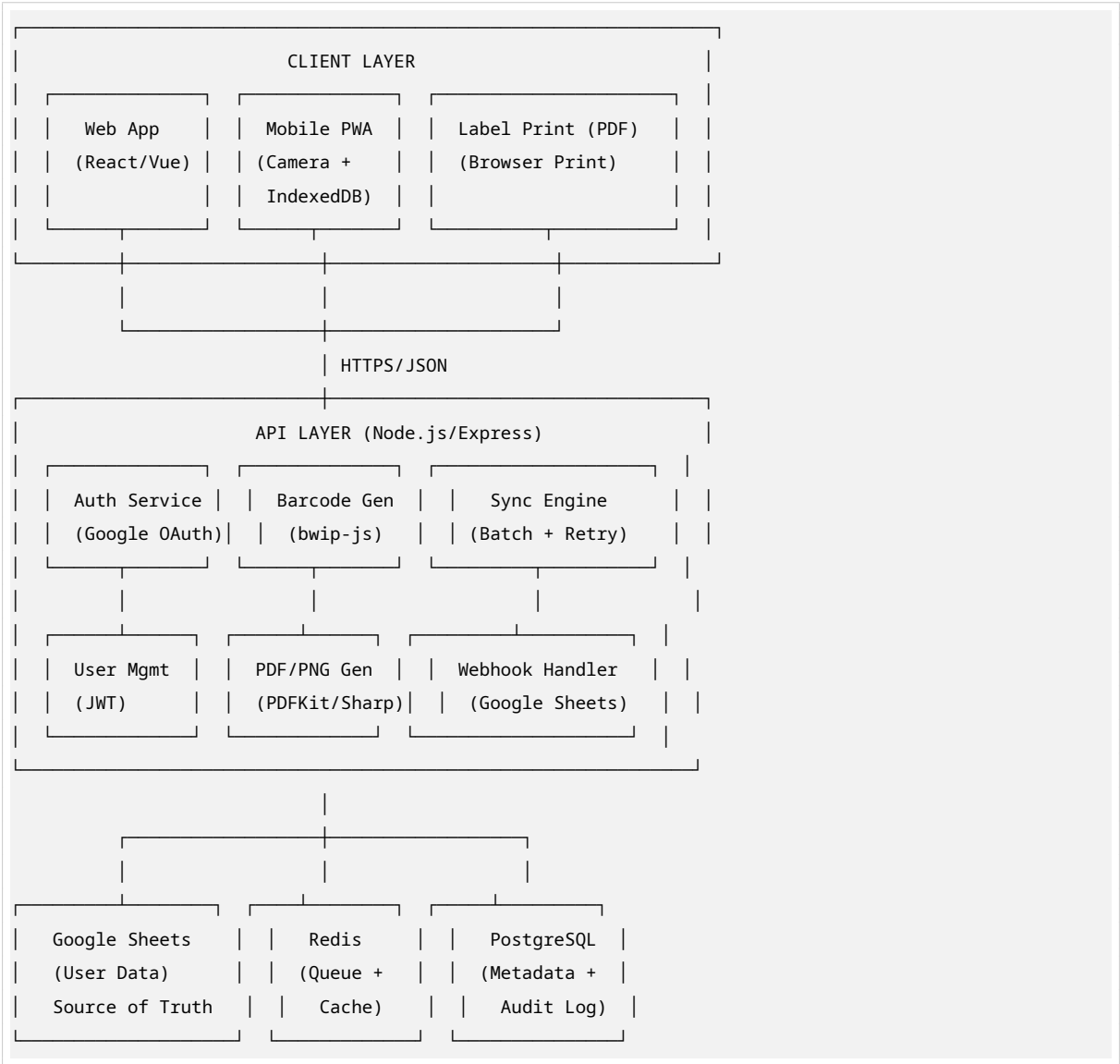
ID	Requirement	Target	Measurement
NFR-11	Setup time (upload to first scan)	<30 min	Unmoderated usability test, n=5 per persona
NFR-12	Task success rate (core flows)	80%	First-attempt completion
NFR-13	System Usability Scale (SUS) score	70	Post-test questionnaire
NFR-14	Mobile PWA install prompt	P1	Chrome/Edge/Safari add-to-home-screen
NFR-15	Accessibility (WCAG 2.1 AA)	P2	Screen reader compatible, color-blind safe

5.4 Reliability

ID	Requirement	Target
NFR-16	Uptime (prototype phase)	99.5%
NFR-17	Google Sheets API error recovery	Auto-retry 3× with backoff
NFR-18	Data backup (user responsibility)	Google Drive version history
NFR-19	graceful degradation (Sheets unavailable)	Queue all operations, notify user

6. Technical Architecture (Prototype)

6.1 System Diagram



6.2 Technology Stack (Prototype)

Layer	Technology	Rationale
Frontend	React 18 + TypeScript + Tailwind CSS	Rapid development, component reuse
Mobile PWA	React + Workbox (service worker)	Offline support, installable
Barcode Generation	bwip-js (Code 128 + QR)	Client-side generation, no server dependency
Backend API	Node.js 20 + Express + TypeScript	Familiar stack, Google API client maturity
Database	PostgreSQL 15 (Neon/Supabase)	User metadata, audit logs, config
Cache/Queue	Redis (Upstash)	Sync batching, rate limit handling
Auth	Google OAuth 2.0 (passport-google-oauth)	Native Sheets integration, SSO

Table 19 – continued

Layer	Technology	Rationale
PDF Generation	PDFKit + Sharp (PNG)	Server-side rendering, thermal printer support
Hosting	Vercel (frontend) + Railway/Render (API)	Rapid deploy, auto-scaling
Monitoring	LogRocket + Sentry	Session replay, error tracking

6.3 Google Sheets API Strategy

Rate Limit Handling:

- Batch writes: 50 rows per batchUpdate request
- Exponential backoff: 200ms → 800ms → 2500ms for 429 errors
- Queue depth monitoring: Alert at >100 pending operations
- Fallback: Queue to Redis if Sheets API unavailable >5 min

Data Model:

User Sheet (Source of Truth):

Item_ID	Name	SKU	Category	Qty	Barcode	Last_Scan_Date
1001	iPhone 15	MOB001	Mobiles	12	BSKB-1001	2026-05-28
1002	Samsung	MOB002	Mobiles	8	BSKB-1002	2026-05-29
...	...	...	...	...	...	...

7. User Interface Requirements

7.1 Key Screens (Prototype)

Screen 1: Dashboard (Post-Setup)

Total Items

523

Scans Today

47

Low Stock

12

Quick Actions:



Upload Sheet



Generate Labels



Open Scanner

Recent Activity:

•

10:23 AM — Koteswara scanned MOB-2847 (Mobiles)

•

10:15 AM — Assistant_2 checked out TV-192 (Rental)

• 09:47 AM — Reorder alert: AC-45 (Qty: 2, Threshold: 5)	
[View Full Inventory] [Generate Report]	

Screen 2: Mobile PWA Scanner

⬅️ BACK

Scan Item

⚡️ 🔧

Camera Viewfinder

Align barcode

 Batch Mode: ON | Count: 23

Last Scan: BSKB-2847 — iPhone 15 Pro
Qty: 12 → 11 | Location: Shelf A3

[Manual Entry] [View History]

Screen 3: Label Print Configuration

⬅️ BACK

Generate Labels — 523 items selected

Format: [PDF ▼] [PNG ▼]

Template: [Avery 5160 ▼] [2"×1" Thermal ▼]

Layout Editor (Drag fields):

[Product Name] [SKU]

[BARCODE IMAGE]

[Location] [Price]

Barcode Format: [Code 128 ●] [QR Code ○]

Prefix: [BSKB-_____] Start: [1001]

 Print Preview

 Generate & Download

8. Data Requirements

8.1 Data Model (Simplified for Prototype)

```
User (PostgreSQL)
├─ id: UUID (PK)
├─ google_id: String (unique)
├─ email: String
├─ name: String
├─ company: String
├─ role: Enum [owner, manager, operator]
├─ created_at: Timestamp
└─ settings: JSONB

InventoryItem (Google Sheets — user-owned)
├─ item_id: String (user-defined or auto)
├─ name: String
├─ sku: String (unique within sheet)
├─ category: String
├─ quantity: Number
├─ location: String
├─ barcode: String (generated)
├─ custom_fields: JSON (prototype: up to 4 keys)
├─ last_scan_date: Timestamp
├─ last_scan_user: String
└─ created_at: Timestamp

ScanEvent (PostgreSQL — audit log)
├─ id: UUID (PK)
├─ user_id: UUID (FK)
├─ item_id: String
├─ action: Enum [check_in, check_out, transfer, adjust]
├─ quantity_delta: Number
├─ from_location: String
├─ to_location: String
├─ timestamp: Timestamp
├─ device_type: String
└─ sync_status: Enum [pending, synced, failed]

LabelTemplate (PostgreSQL)
├─ id: UUID (PK)
├─ user_id: UUID (FK)
├─ name: String
├─ format: Enum [pdf_a4, pdf_letter, png_2x1, png_4x6]
├─ layout_config: JSONB (field positions, fonts)
└─ created_at: Timestamp
```

8.2 Google Sheets Schema (Auto-Generated)

Column	Type	Source	Example
A	Item_ID	User or Auto	1001
B	Name	User	iPhone 15 Pro 128GB
C	SKU	User	MOB-2847
D	Category	User	Mobiles
E	Quantity	User/App	12
F	Location	User/App	Warehouse A, Shelf 3
G	Barcode	App-Generated	BSKB-1001
H	Last_Scan_Date	App-Generated	2026-05-28T10:23:00Z
I	Last_Scan_User	App-Generated	koteswara@bskb.com
J	Reorder_Point	User	5
K	Custom_Field_1	User	Supplier: Apple India
L	Custom_Field_2	User	Cost_Price: 72,000

9. Integration Requirements

9.1 Google Workspace (P0)

Integration	Scope	Use Case
Google Sheets API (v4)	drive.file + spreadsheets	Read/write inventory data, create new sheets
Google Drive API (v3)	drive.file	Store generated labels, backup exports
Google OAuth 2.0	openid email profile	Authentication, SSO, user profile

9.2 Prototype-Only Integrations (P2)

Integration	Use Case	Validation Goal
Slack Webhook	Scan notifications to team channel	SS segment team adoption
Email (SendGrid)	Reorder alerts, overdue notices	AB segment expiration alerts
Zapier (Trigger)	Export to QuickBooks/Xero	KD segment accountant sharing

10. Analytics & Metrics

10.1 Prototype Success Metrics (6-Week Validation)

Metric	Target	Measurement Method	Persona Focus
Setup Completion Rate	80%	Funnel analysis (signup → upload → generate → first scan)	All
Time to First Scan	<30 min	Session recording (LogRocket)	KD, SS
Daily Active Users (DAU)	60% of registered	Backend analytics	All
Scan Accuracy	95%	Manual validation of 100 random scans	All
Google Sheets Sync Success	99%	Sync status dashboard	All
NPS Score	40	In-app survey (week 2, 4, 6)	All
Discrepancy	1 per first count	User-reported + system-detected	KD, IP, AB
Discovery Rate			
Feature Request Volume	<5 critical per persona	Support tickets + interviews	All

10.2 Leading Indicators (Weekly)

Indicator	Target	Action if Below Target
Upload success rate	90%	Simplify column mapping, add templates
Barcode generation completion	95%	Optimize batch processing, add progress
PWA install rate	30%	Improve install prompt, add value messaging
Offline scan queue (avg)	<50 pending	Improve sync reliability, add retry
Support ticket volume	<2 per 10 users	Improve onboarding, add contextual help

11. Risk Analysis & Mitigation

Risk	Probability	Impact	Mitigation Strategy
Google Sheets API rate limits block scaling	High	Critical	Implement Redis queue, batch writes, exponential backoff
Users fear data lock-in (KD, AB)	High	High	Emphasize Google Drive ownership, provide CSV export, clear privacy policy
Thermal printer label alignment issues	Medium	Medium	Print calibration wizard, community-tested presets

Table 25 – continued

Risk	Probability	Impact	Mitigation Strategy
Mobile camera scan failures in poor light	Medium	Medium	Torch control, manual entry fallback, scan quality feedback
Enterprise procurement delays (IP)	High	Medium	Pre-packaged pilot program with 30-day success metrics
Public sector IT review blocks (AB)	High	High	Pre-built compliance documentation, HIPAA/FIPPA FAQ
Team adoption resistance (SS, IP)	Medium	High	Gamification (leaderboard), micro-training videos, Slack integration
Competitor response (Sortly, Asset Panda)	Low	Medium	Speed to market with Sheets integration, price anchoring at \$39

12. Prototype Timeline (6 Weeks)

Week 1-2: Foundation

- Google OAuth + user authentication
- Google Sheets API connection (read/write)
- Basic inventory upload (CSV + Sheets)
- PostgreSQL schema + user metadata storage

Week 3-4: Core Features

- Code 128 + QR Code generation (bwip-js)
- PDF label generation (Avery 5160 template)
- Mobile PWA scanner (camera + barcode detection)
- Check-in/check-out + location transfer
- Basic offline queue + sync

Week 5: Persona-Specific Features

- Custom fields (IP scenario)
- Expiration alerts (AB scenario)
- Team multi-user + due-dates (SS scenario)
- Multi-category support (KD scenario)

Week 6: Polish & Pilot Prep

- Print calibration wizard
- Onboarding tutorial (3-step interactive)
- Error handling + user feedback
- Analytics instrumentation
- Beta user recruitment (2-3 per segment)

Week 7-9: Pilot & Iterate

- Daily standup with beta users
- Weekly metric review
- Rapid iteration on blocking issues
- NPS survey + exit interviews
- Go/No-Go decision for full product

13. Open Questions

1. **Barcode format prioritization:** Should QR Code be default for SS/AB segments despite Code 128 being more compact for KD/IP?
2. **Offline sync conflict resolution:** Is “last write wins” acceptable for prototype, or do we need basic versioning?
3. **ZPL support scope:** Should prototype include ZPL for IP’s Zebra printers, or defer to full product?
4. **HIPAA compliance:** Can we leverage Google’s BAA for prototype, or do we need separate compliance review?
5. **Pricing validation:** Is \$39/month validated by prototype adoption, or do we need pricing experiments?

14. Appendices

Appendix A: Persona Reference Documents

- [User Persona Cards \(PDF\)](#)
- [Empathy Maps \(PDF\)](#)
- [User Journey Maps \(PDF\)](#)

Appendix B: Competitive Analysis Summary

Competitor	Price	Strength	Weakness	Our Differentiator
Sortly	\$39/mo	Beautiful UI	No Sheets integration	Native Sheets sync
inFlow	\$89/mo	Full ERP features	Complex setup, expensive	30-min setup, \$39 price
Asset Panda	\$48/asset/yr	Enterprise features	Per-asset pricing, complex	Unlimited items, simple

Table 26 – continued

Competitor	Price	Strength	Weakness	Our Differentiator
EZOfficeInventory	\$35/mo	Asset-focused	No retail/SMB focus	Multi-segment support

Appendix C: Glossary

- **SKU:** Stock Keeping Unit — unique item identifier
- **PWA:** Progressive Web App — installable web application
- **ZPL:** Zebra Programming Language — thermal printer command language
- **VFC:** Vaccines for Children — US federal vaccine program
- **BAA:** Business Associate Agreement — HIPAA compliance contract
- **CapEx:** Capital Expenditure — long-term asset investment

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This PRD is based on real user data from LinkedIn profiles, market research conducted May 2026, and competitive analysis of the barcode inventory management sector. All persona details are derived from publicly available professional information and used for product strategy purposes.